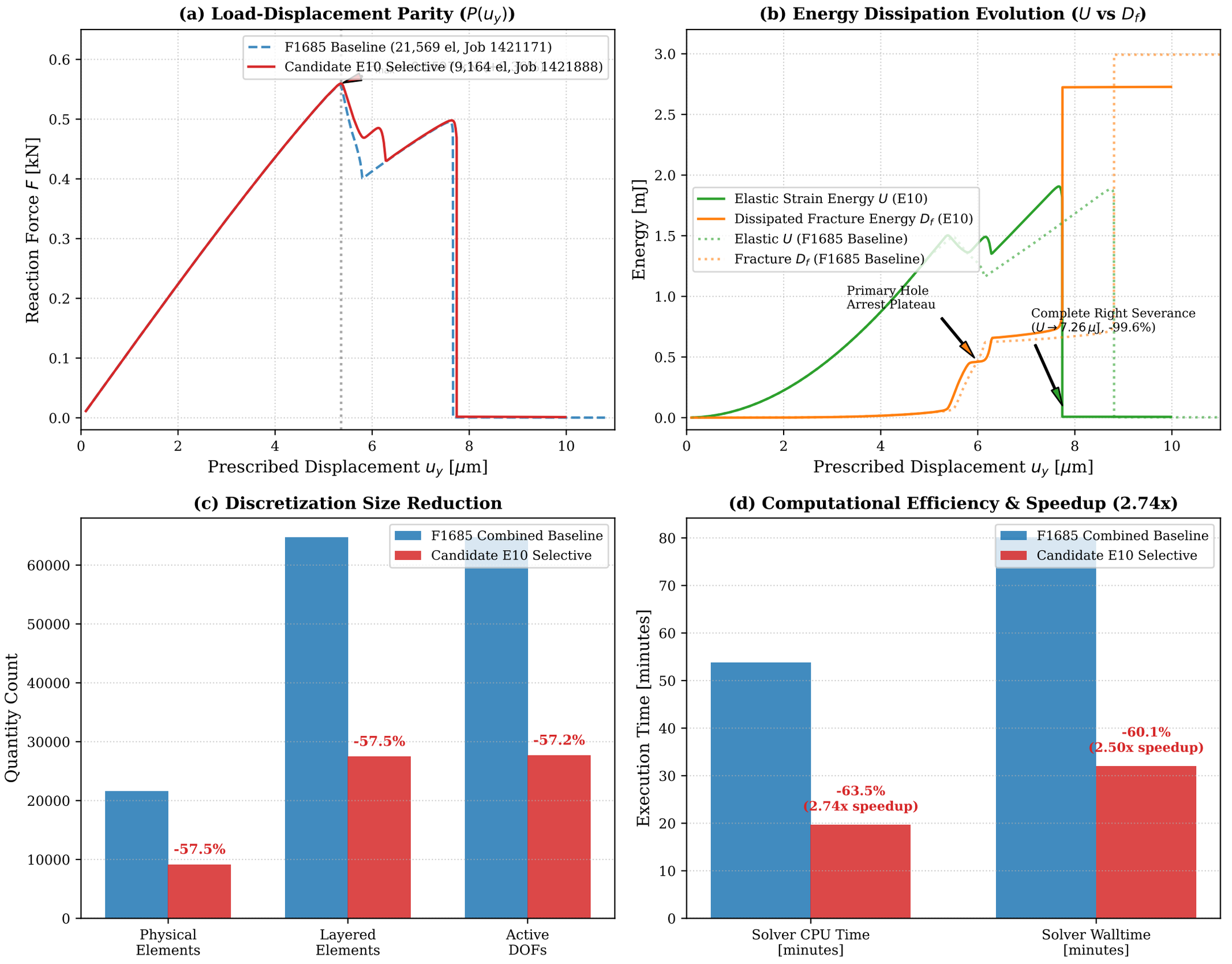
(a) Load-Displacement Parity ($P(u_y)$)
F1685 Baseline (21,569 el, Job 1421171)
Candidate E10 Selective (9,164 el, Job 1421888)
Reaction Force F [kN]
Prescribed Displacement u_y [μm]
(b) Energy Dissipation Evolution (U vs D_f)
Elastic Strain Energy U (E10)
Dissipated Fracture Energy D_f (E10)
Elastic U (F1685 Baseline)
Fracture D_f (F1685 Baseline)
Primary Hole
Arrest Plateau
Complete Right Severance
($U \rightarrow 7.26\,\mu$J, -99.6%)
Energy [mJ]
Prescribed Displacement u_y [μm]
(c) Discretization Size Reduction
F1685 Combined Baseline
Candidate E10 Selective
-57.5%
-57.5%
-57.2%
Quantity Count
Physical Elements
Layered Elements
Active DOFs
(d) Computational Efficiency & Speedup (2.74x)
F1685 Combined Baseline
Candidate E10 Selective
-63.5%
(2.74x speedup)
-60.1%
(2.50x speedup)
Execution Time [minutes]
Solver CPU Time [minutes]
Solver Walltime [minutes]